EY

HULT INTERNATIONAL BUSINESS SCHOOL

Predicting Urban Heat: A Machine Learning Approach Across Three Cities

Predicting UHI with Satellite Data

Business Challenge II
Team 4: Carolina Trovisco, Filippo Beni, João Ponte, Mickias Ambaye, Youness Yachruti, Yousra Sajjad

Executive Summary

Urban Heat Islands (UHI) are urban areas that are warmer than surrounding areas, with local-scale temperature differences exceeding 10°C due to dense building layouts trapping heat, impervious surface heat absorption, and industry/transportation waste heat.

Key Findings: Santiago's heat is shaped by elevation and terrain interactions, while Rio's is driven by thermal surface absorption and building density. The combined model transfers these signals to predict UHI intensity in Freetown.

Recommendations: Our models inform three cross-city recommendations: prioritizing reflective urban materials, improving street-level ventilation and shading, and tailoring implementation to each city's unique density, planning capacity, and economic reality.

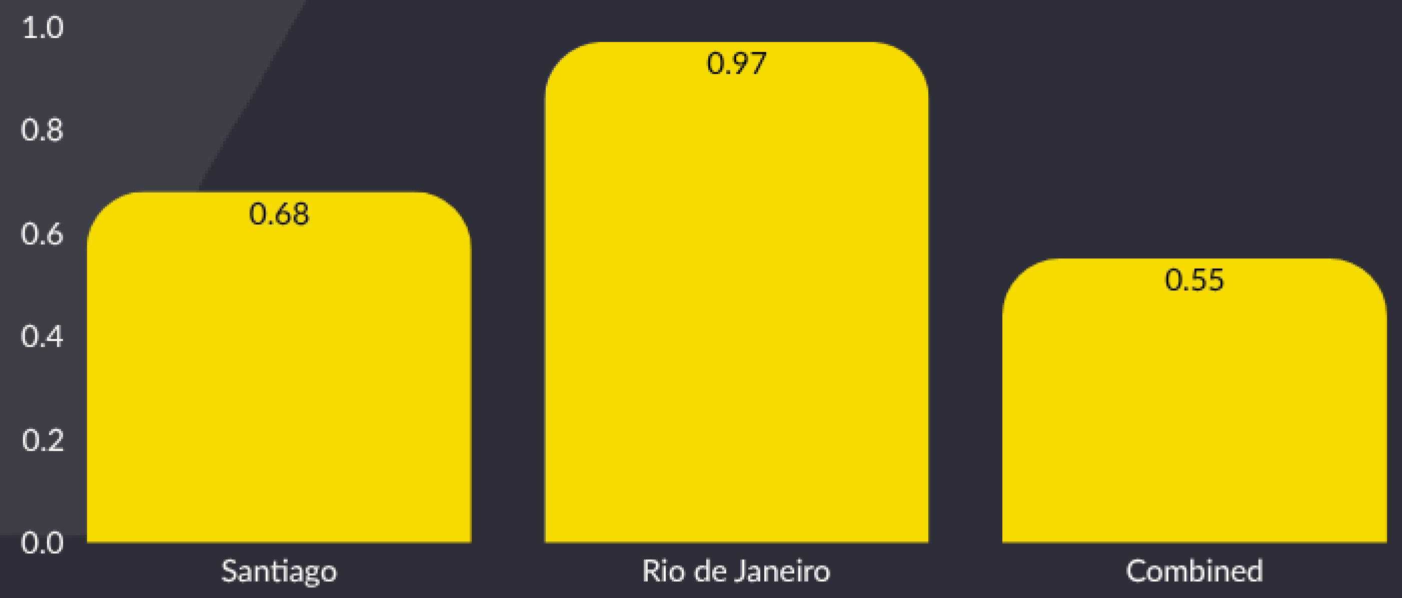
Data Points 50,150
Santiago (21,662) & Rio (28,488)

Satellite Indices: 25
Sentinel-2 spectral indices across vegetation, water, built-up & surface categories

Engineered Features 6
Interaction features (e.g. LST × NDVI, elevation × LST)

Thermal & Terrain: 2
Landsat LST + Elevation

Overall Model Performances (F1 Score)



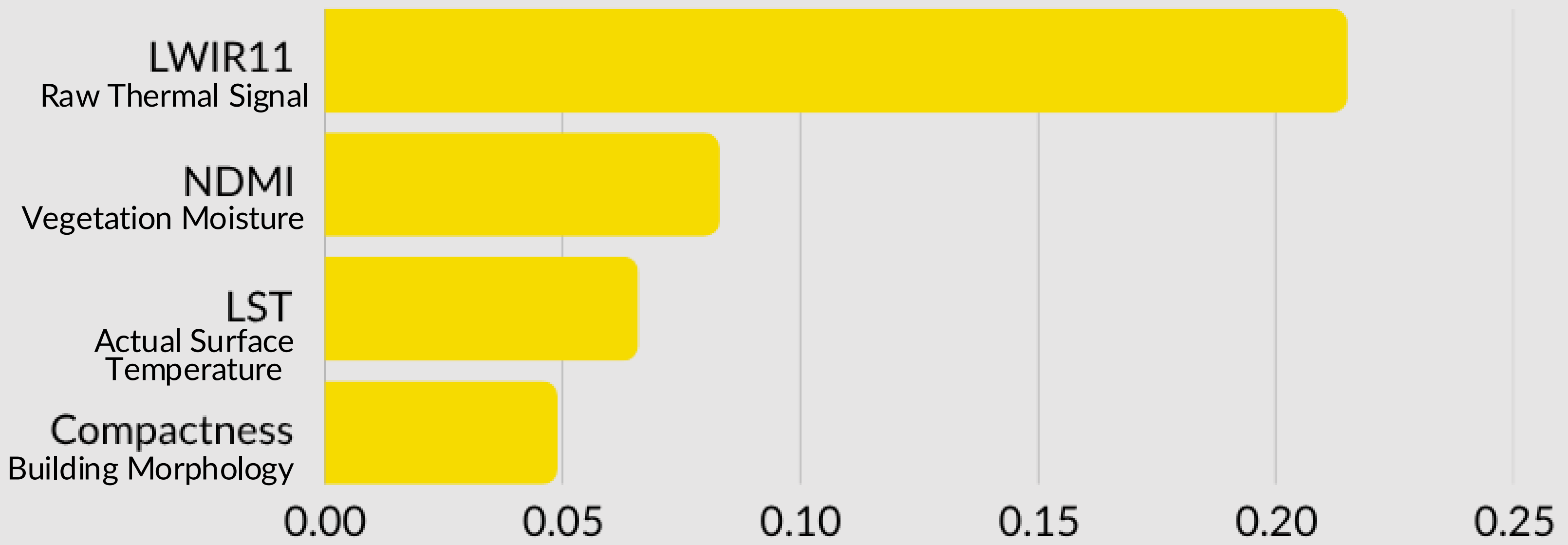
For further details, refer to appendix slide number 12-14

Rio de Janeiro, Brazil Model

MODEL PERFORMANCE

0.959	0.960	0.959
F1 Score	Precision	Recall

FEATURE OF IMPORTANCE



Best Model

- XGBoost

Key Drawbacks

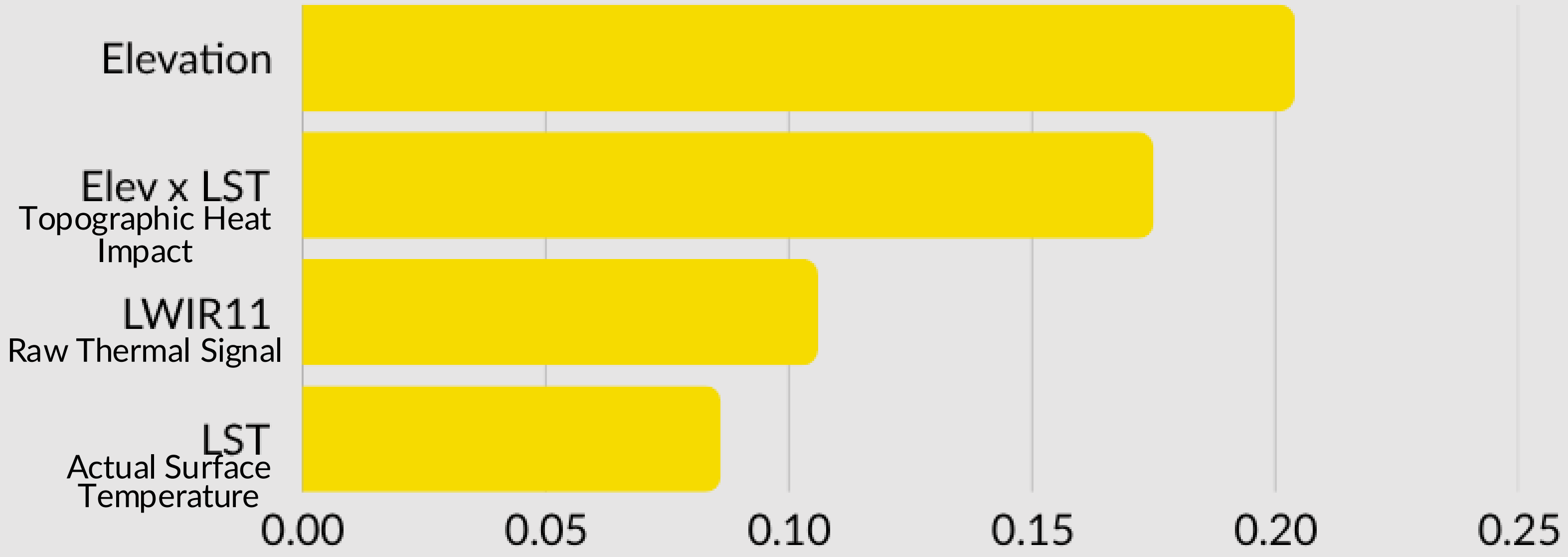
- Data Quality SCL (Scene Classification Layer) cloud filtering was applied to address approximately 50% cloud coverage detected in one satellite scene, with median compositing across multiple scenes used to ensure clean and reliable data extraction.

Santiago, Chile Model

MODEL PERFORMANCE

0.690	0.694	0.690
F1 Score	Precision	Recall

FEATURE OF IMPORTANCE



Best Model

- Random Forest Classifier

Key Drawbacks

- Balancing bias and variance required careful hyperparameter tuning to find the optimal model fit, with both the weighted F1 score and generalization gap used as dual indicators to ensure the model was neither overfitting nor underfitting across cities.

Does a Combined Model predict Freetown?

MODEL PERFORMANCE

0.58	0.59	0.58
F1 Score	Precision	Recall

OVERALL FEATURE OF IMPORTANCE

Best Model

- Ensemble of XGBoost and Random Forest

Key Drawbacks

- Isolating UHI-relevant signals from city-specific characteristics in both Brazil and Chile proved challenging, as raw features encoded location identity rather than heat patterns (Quantile Transformation and PCA)
- Given the ambiguity between the three UHI classifications, separate models were required for each class to ensure the boundaries between High, Medium, and Low were learned independently and with greater precision.

Key Insights

1. Urban heat drivers are city-specific, not universal

The heat is different as it is dependent on the city's terrain, land cover and urban structure

2. Santiago's heat pattern is more structurally complex to model

The heat is shaped by elevation, slope, and built form, making hotspot boundaries less clean and harder for the model to separate

3. Rio's building materials make it easier to detect UHI

The UHI is clearer due to intense year-round solar exposure combines with high shares of concrete, asphalt, and other impervious materials that absorb and retain heat quickly

4. Rooftop materials and color likely amplified the UHI of Freetown, while making it harder to classify

LWIR11, LST, and Blue together suggest that Freetown's roof materials and colors both intensified surface heating and blurred clean separation between UHI classes.

Recommendations

1

Deploy reflective materials where compactness scores are highest

Rio's XGBoost model ranked building compactness as a top 4 predictor. The model's compactness x LST scores can be used to rank neighborhoods by intervention priority. In Freetown, the Blue band signal points directly to dark roof colors as heat amplifiers. A satellite-guided roof audit in high-LWIR11 zones is a low-cost, high-impact first step.

2

Use elevation x LST interaction to locate ventilation corridors in Santiago

Santiago's model found elevation and its interaction with LST to be the two strongest features, more predictive than raw temperature alone. The engineered LST elev resid feature identifies which streets sit in thermal trap zones. Ventilation corridors and tree canopy placement should be mandated there first. This is a surgical intervention, and Santiago's model already did the targeting.

3

Institutionalize the satellite pipeline as a quarterly monitoring tool

The most underused asset from this project is the pipeline itself: Sentinel-2 and Landsat, cloud-masked, 100m resolution, 25 indices, UHI classification, repeatable at near-zero cost via Microsoft Planetary Computer. Cities should run this quarterly so emerging hotspots are flagged before they become crises. For Freetown, where ground truth is absent and settlements shift fast, this monitoring layer is more valuable than any static intervention plan.

Key Challenges & Our Path to Stability

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- API & Data Access Rate-limited and paid APIs slowed data extraction
- Data Granularity
- Cross-City Standardization
- Cross-City Validation & Model Instability
- Blind Prediction on Sierra Leone
- Class Confusion (High / Medium / Low)
- Validating Generalization to Sierra Leone

- Free API & Caching
- Resolution Testing
- Pixel-Level Cloud Masking
- Per-City Quantile Normalization and PCA
- Leave-One-City-Out Validation
- QuantileTransformer + Balanced Weighting
- K Mean Clusturing

For further details, refer to appendix slide number 15 - 17

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Our Team

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Thank you

Open for Questions

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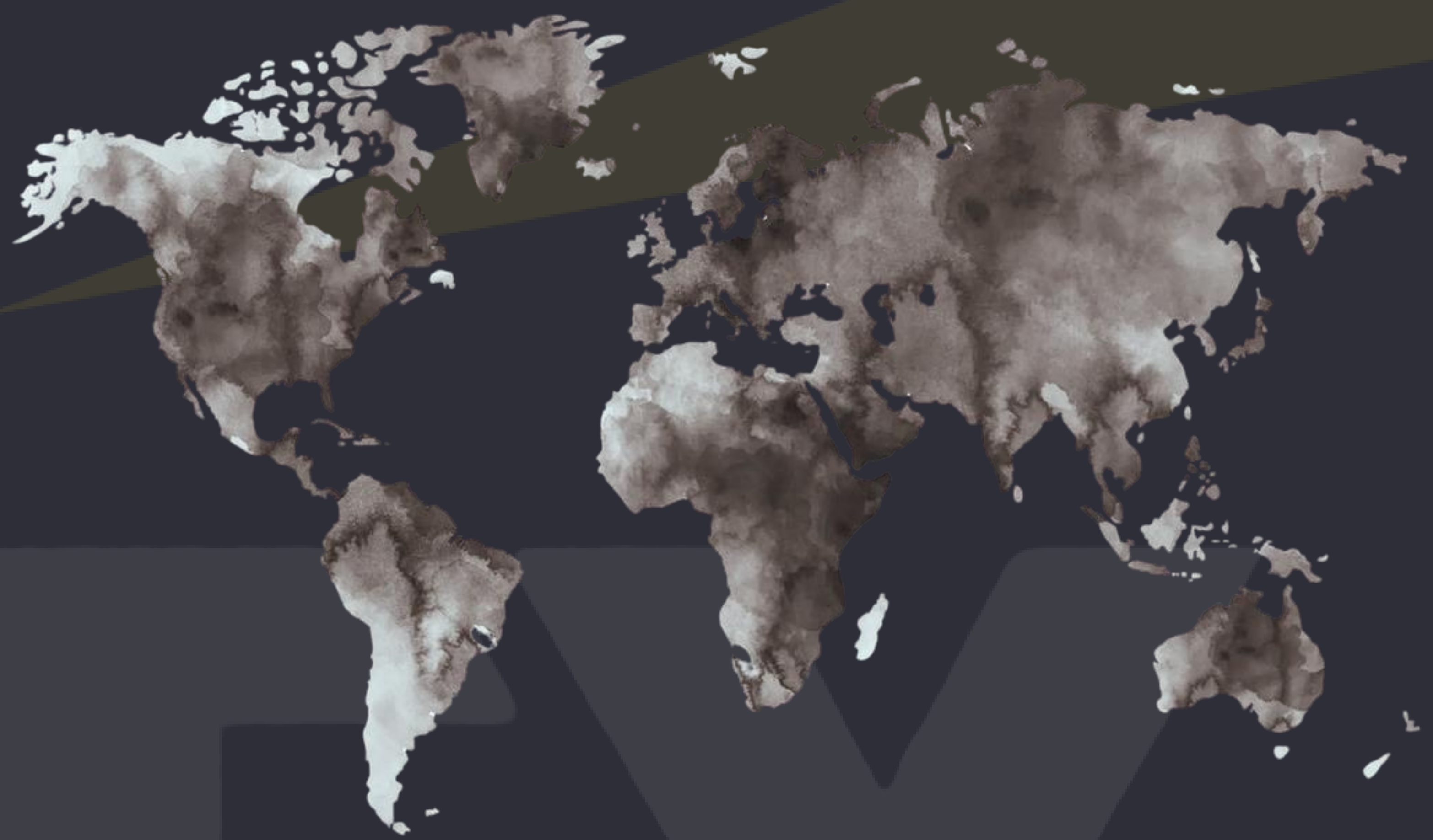
Appendix - Cross-City Applications



- Models are trained on one labeled city and then tested on the another entirely untested city, which confirms that the model learns UHI patterns.
- Per-city Quantile Normalization converts absolute temperature values into relative percentile ranks, ensuring that the high UHI signal in a mountain city transfers to a tropical coastal one without issues or bias.

Appendix - Features Which Have Been Created

Feature	Formula	Source Bands	Why
LST × NDVI	LST × NDVI	Landsat B10, S2 B08, B04	Hot bare vs. cool vegetated
LST × NDBI	LST × NDBI	Landsat B10, S2 B11, B08	Heat amplified by built-up density
elev × LST	elevation × LST	Copernicus DEM, Landsat B10	Topographic heat amplification effect
NDVI – NDBI	NDVI – NDBI	S2 B08, B04, B11	Vegetation versus urban development contrast
LST × Albedo	LST × Albedo	Landsat B10, S2 B02, B04, B08, B11, B12	Reflectivity coupled with surface temperature
BU × LST	(NDBI – NDVI) × LST	Landsat B10, S2 B11, B08, B04	Urban footprint weighted by heat

Feature	Category	Formula	Source Band(s)	Why
NDVI	Vegetation	(B08 − B04) / (B08 + B04)	B08, B04	Vegetation cools urban heat areas
EVI	Vegetation	2.5 × (B08 − B04) / (B08 + 6×B04 − 7.5×B02 + 10000)	B08, B04, B02	Sensitive vegetation in dense canopy
SAVI	Vegetation	(B08 − B04) / (B08 + B04 + 0.5) × 1.5	B08, B04	Adjusts vegetation for bare soil
GNDVI	Vegetation	(B08 − B03) / (B08 + B03)	B08, B03	Alternate vegetation using green band
LAI	Vegetation	0.57 × exp(2.33 × NDVI)	Derived from NDVI	Canopy density from vegetation estimate
NDWI	Water	(B03 − B08) / (B03 + B08)	B03, B08	Detects water bodies, cooling effect
MNDWI	Water	(B03 − B11) / (B03 + B11)	B03, B11	Separates water from urban surfaces
NDMI	Water / Moisture	(B08 − B11) / (B08 + B11)	B08, B11	Vegetation moisture reduces heat absorption
NDBI	Built-up	(B11 − B08) / (B11 + B08)	B11, B08	Urban surfaces retain more heat
BU	Built-up	NDBI − NDVI	Derived	Direct urban vs. vegetation contrast
ISA	Built-up	(NDBI − NDVI) / (NDBI + NDVI)	Derived	Impervious surface fraction estimate
NDISI	Built-up	(B11 − (B02+B08+B12)/3) / (B11 + (B02+B08+B12)/3)	B11, B02, B08, B12	Isolates impervious from soil/vegetation
NDBaI	Soil / Bare	(B11 − B12) / (B11 + B12)	B11, B12	Identifies exposed bare ground areas

Feature	Category	Formula	Source Band(s)	Why
BSI	Soil / Bare	((B11+B04) − (B08+B02)) / ((B11+B04) + (B08+B02))	B11, B04, B08, B02	Bare soil heats faster than vegetation
DBSI	Soil / Bare	(B11 − B03) / (B11 + B03) − NDVI	B11, B03, NDVI	Dry bare soil drives surface heating
LSE	Surface Energy	0.004 × Pv + 0.986	Derived from NDVI	Controls how surfaces radiate heat
Albedo	Surface Energy	0.356×B02 + 0.130×B04 + 0.373×B08 + 0.085×B11 + 0.072×B12 − 0.0018	B02, B04, B08, B11, B12	Reflectivity determines surface heat absorption
SWIR1_NIR	Band Ratio	B11 / B08	B11, B08	Urban texture from infrared contrast
SWIR2_NIR	Band Ratio	B12 / B08	B12, B08	Secondary infrared urban surface ratio
LST	Thermal	BT / (1 + (λ × BT / ρ) × ln(ε))	Landsat B10 (lwir11)	Actual surface temperature, UHI proxy
lwir11	Thermal	Raw brightness temperature from Landsat thermal band	Landsat B10	Raw thermal signal before correction
NDVI_Landsat	Vegetation	(B5 − B4) / (B5 + B4)	Landsat B5, B4	Independent vegetation from different sensor
Elevation	Topographic	Extracted directly	Copernicus DEM	Height above sea level, temperature driver
LST_elev_resid	Topographic	LST − predicted LST from elevation regression	Derived	Temperature after removing elevation effect
building_density_100m	Structural	Count of buildings / buffer area (m²)	3D-GloBFP shapefile	Building count per urban area unit

Appendix - Key Challenges

API & Data Access Rate-limited and paid APIs slowed data extraction

- Cloud-covered satellite imagery required masking and resourcing cleaner scenes

Data Granularity

- Some external datasets lacked the granular spatial detail such as humidity needed for modelling and had to be discarded entirely

Cross-City Standardization

- A "High" temperature in Santiago ≠ "High" temperature in Rio and city-level normalization was required to make temperature labels comparable across locations

Cross-City Validation & Model Instability

- Training on one city and testing on another revealed generalization gaps as high as 0.6 in some models, making stability a core challenge

Blind Prediction on Sierra Leone

- With no ground-truth labels available, validating model performance on Freetown relied entirely on internal consistency and cross-city transfer learning

Class Confusion (High / Medium / Low)

- The Medium UHI class sits in an ambiguous spectral zone, making it consistently the hardest class for models to correctly distinguish

Validating Generalization to Sierra Leone

- With no ground-truth labels for Freetown, cross-city validation alone could not confirm the model would generalize to an entirely unseen city because passing Chile/Brazil cross-validation did not guarantee transfer to a different climate and urban environment

Appendix - Our Path to Stability

Free API & Caching

- Switched to Microsoft Planetary Computer's free STAC API and built a parquet caching layer to avoid redundant downloads

Resolution Testing

- Tested multiple spatial resolutions and validated 100m with a 5×5 pixel bounding box as the optimal balance between detail and noise

Pixel-Level Cloud Masking

- Replaced single-scene extraction with SCL and QA_PIXEL bit masking combined with median compositing across multiple scenes which eliminating data gaps entirely

Per-City Quantile Normalization + PCA

- Converted absolute feature values into within-city percentile ranks so that "hot for Santiago" and "hot for Rio" carry the same meaning to the model

Leave-One-City-Out Validation

- Adopted cross-validation and reduced the feature set from 12+ correlated variables to 5–6 clean, non-redundant ones to close generalization gaps

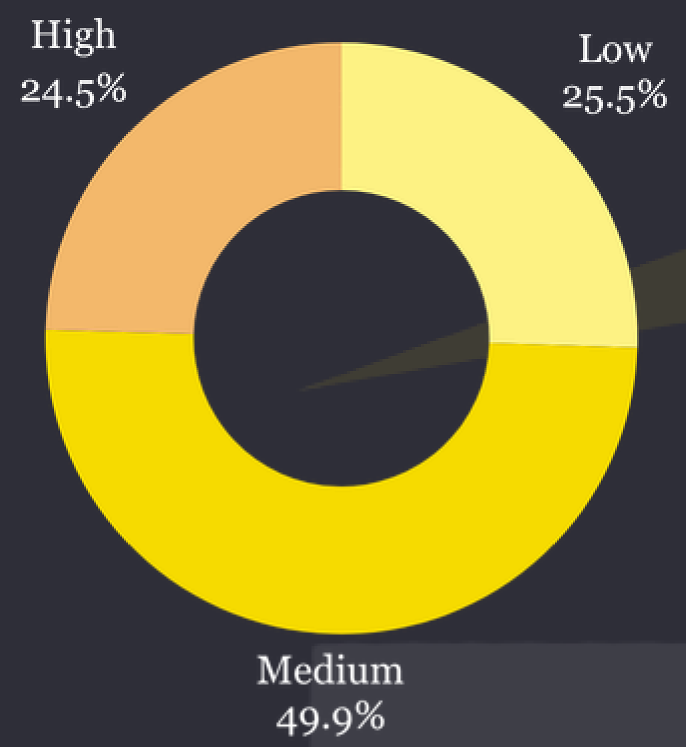
QuantileTransformer + Balanced Weighting

- Combined QuantileTransformer with class_weight="balanced" in Random Forest to spread overlapping Medium class distributions and sharpen class boundaries

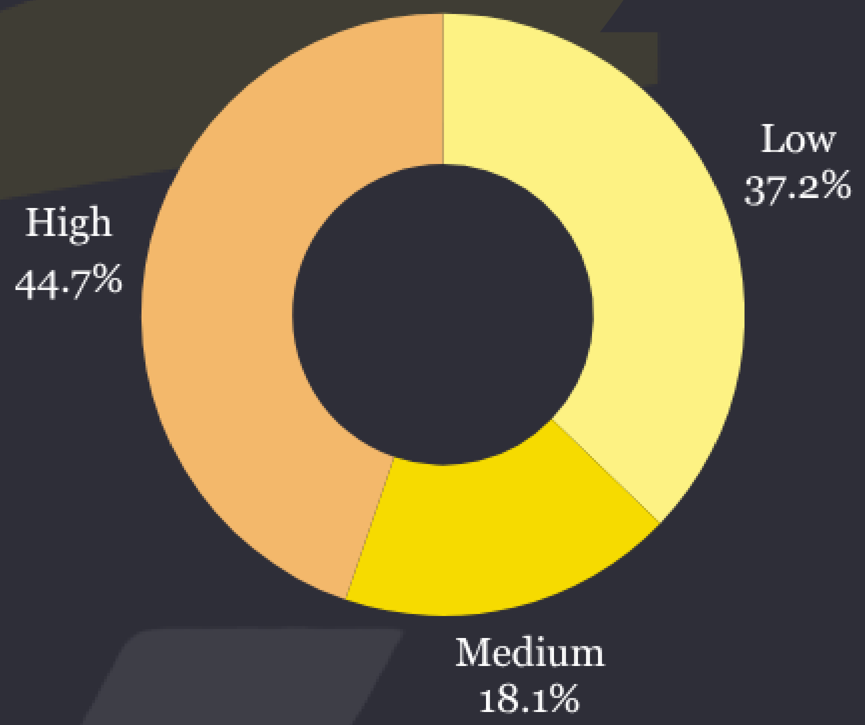
K Mean Synthetic Validation

- Set Built a K Mean Clusturing synthetic Freetown dataset to enable proper train/test splits and hyperparameter tuning against a Sierra Leone-representative distribution

Appendix - Classification Distribution of UHI Classes



Chile



Brazil

Despite the unequal class distribution of low, medium, and high distribution, different measures were undertaken to ensure that one class was not dominating the classes when testing for Freetown.

- Features were grouped together by type; spectral bands together, building information; and normalized using a Quantile Transformer, which maps the group to a 0 to 1 scale.
- PCA is done to effectively stripping city-specific values from the features to reduce the dominant class.
- Class_weight=balanced is also applied to prevent further classes from overshadowing the others when training